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In [5]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

%matplotlib inline

df = pd.read_csv('mall_customers.csv')

df.info()
df.head()
sns.pairplot(df)
plt.show()
features = df.iloc[:, [3, 4]].values
from sklearn.cluster import KMeans

model = KMeans(n_clusters=5, random_state=42, n_init=10)
model.fit(features)

KMeans(n_clusters=5, random_state=42, n_init=10)
Final = df.iloc[:, [3, 4]].copy()

Final['label'] = model.predict(features)

Final.head()
sns.set_style("whitegrid")

sns.FacetGrid(Final, hue="label", height=8) \
    .map(plt.scatter, "Annual Income (k$)", "Spending Score (1-100)") \
    .add_legend()

plt.show()
features_el = df.iloc[:, [2, 3, 4]].values

wcss = []

for i in range(1, 10):
    model = KMeans(
        n_clusters=i,
        random_state=42,
        n_init=10
    )
    model.fit(features_el)
    wcss.append(model.inertia_)

plt.plot(range(1, 10), wcss, marker='o')
plt.title('The Elbow Method')
plt.xlabel('Number of Clusters')
plt.ylabel('WCSS')
plt.show()
```

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<class 'pandas.DataFrame'>
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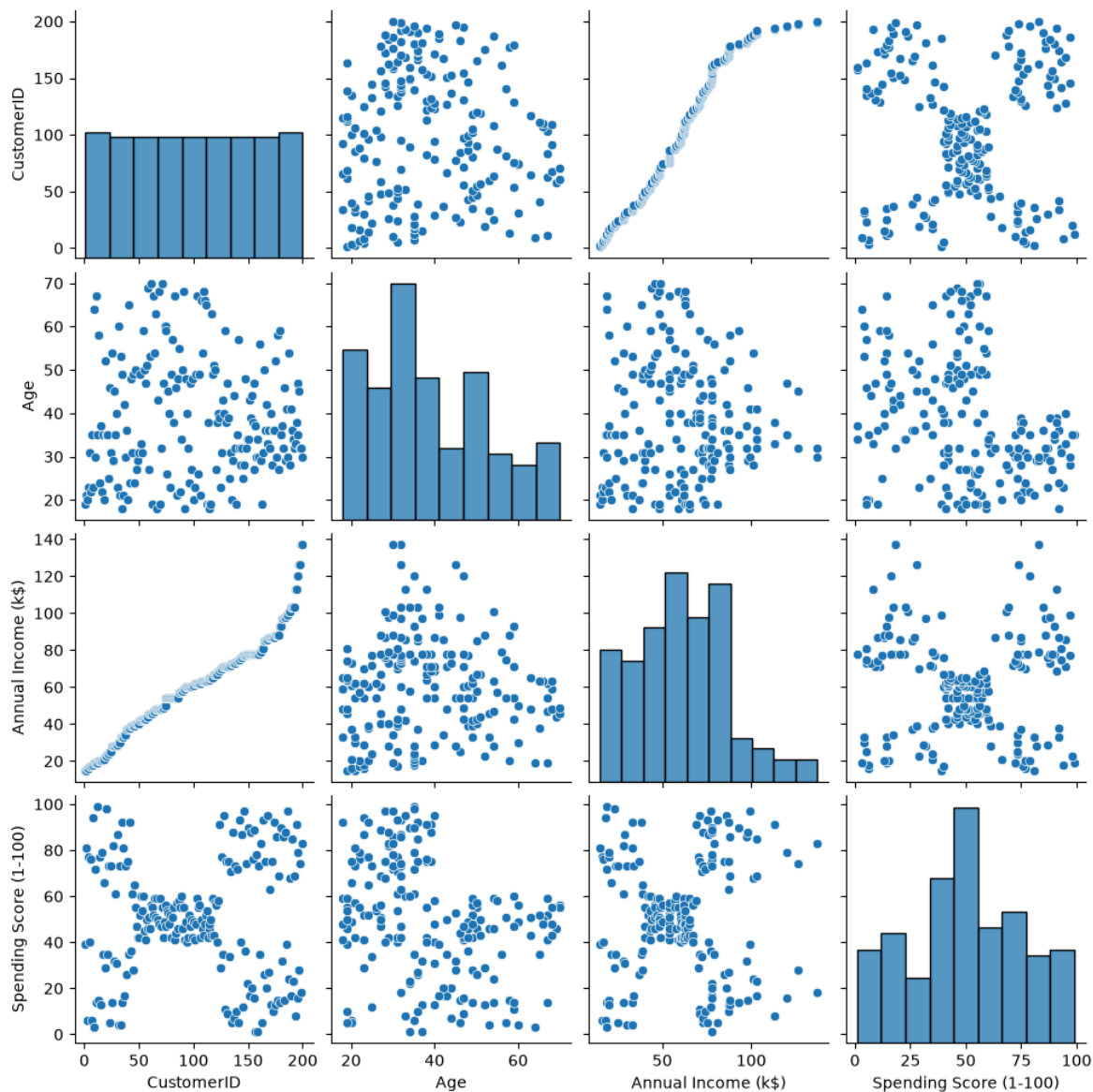
```
RangeIndex: 200 entries, 0 to 199
```

```
Data columns (total 5 columns):
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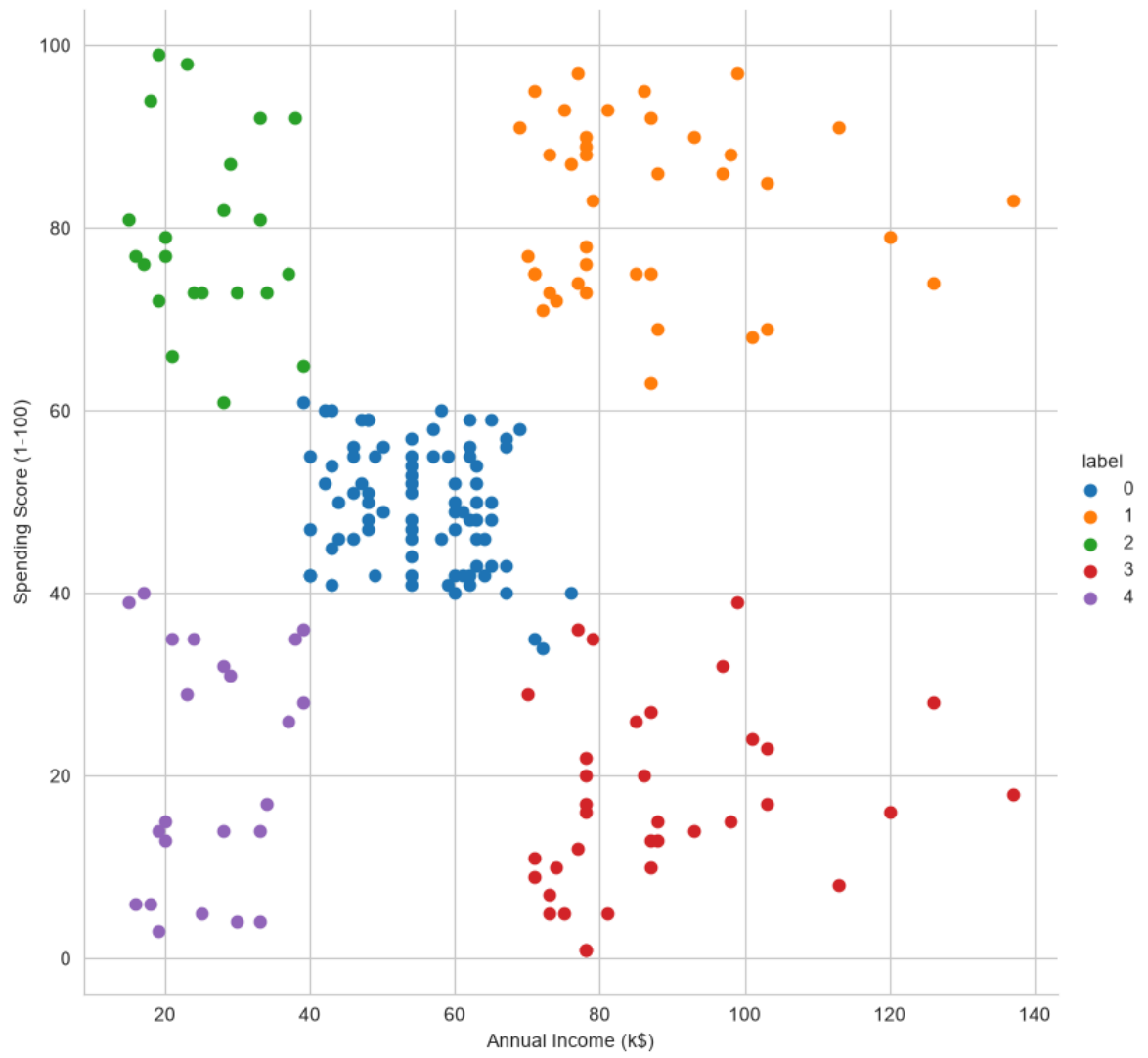
#	Column	Non-Null Count	Dtype
0	CustomerID	200 non-null	int64
1	Gender	200 non-null	str
2	Age	200 non-null	int64
3	Annual Income (k\$)	200 non-null	int64
4	Spending Score (1-100)	200 non-null	int64

```
dtypes: int64(4), str(1)
```

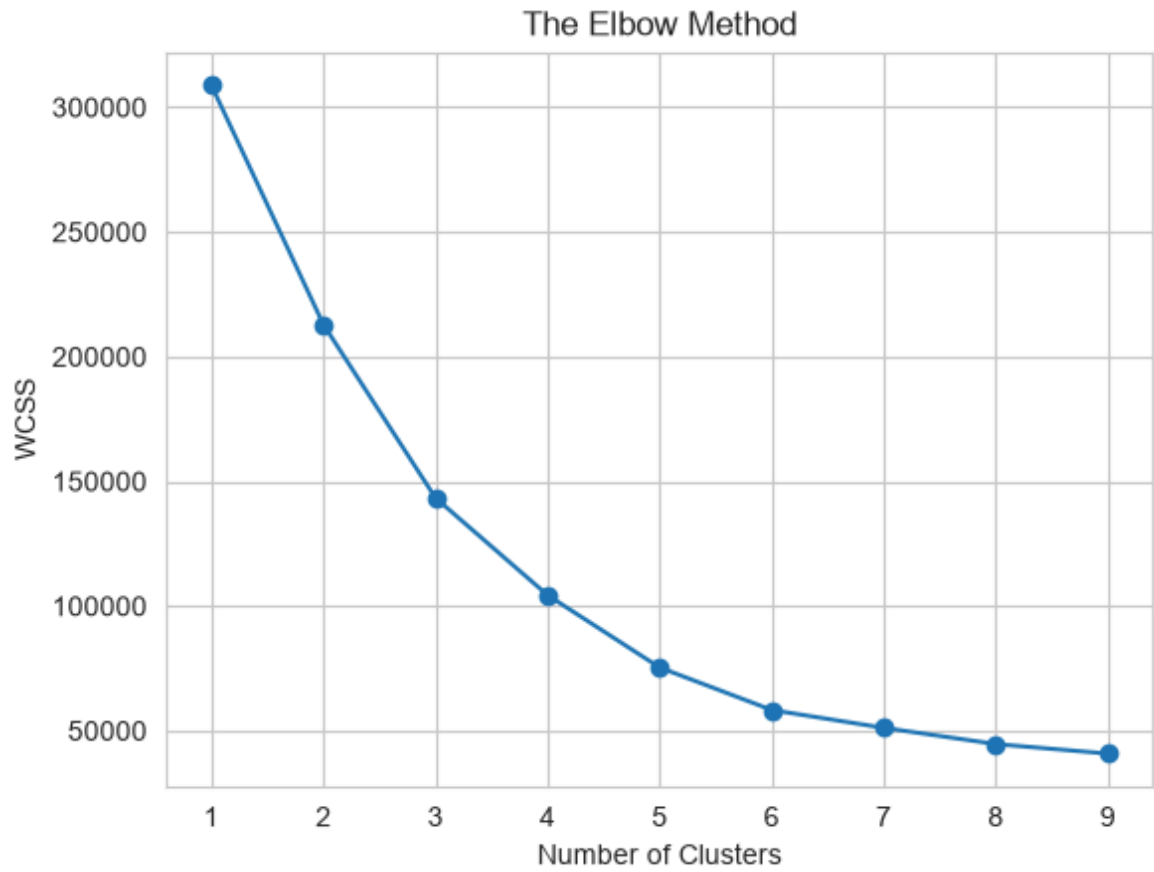
```
memory usage: 8.9 KB
```



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C:\Users\srenaath\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1428: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=1.
  warnings.warn(
```



[illegible]



In []: